

Total No. of Questions : 4]

SEAT No. :

PE-241

[Total No. of Pages : 2

[6580]-602

B.E. (Information Technology) (Insem)

DEEP LEARNING

(2019 Pattern) (Semester - VII) (414443)

Time : 1 Hour]

[Max. Marks : 30

Instructions to the candidates:

- 1) Answer Q.1 or Q.2, Q.3 or Q.4
- 2) Assume suitable data, if Necessary.
- 3) Neat diagrams must be drawn wherever necessary.
- 4) Figures to the right indicate full marks.

Q1) a) Draw & explain diagram of an artificial neuron in a multilayer perceptron neural network. Write mathematical expressions for all activation functions. [5]

b) Explain in detail the concept of *mini-batch* Differentiate between SGD and *mini-batch*. [5]

c) Why regularization is required? Differentiate between L1 and L2 regularization. [5]

OR

Q2) a) Define the term hyperparameter and list all hyperparameters and their usage range of values with units. [5]

b) What is a gradient? Explain mathematically the conditions for vanishing or exploding gradient? [5]

c) List commonly used optimization algorithms. Explain any one in detail with suitable diagrams. [5]

P.T.O.

Q3) a) Explain layered architecture of Alex-Net with relevant details. **[10]**

b) What are the types of pooling? Consider a simple 4×4 feature map and apply a 2×2 max pooling operation with a stride of 2. Show output feature map. **[5]**

OR

Q4) a) What is parameter sharing related to CNN? How is it done? **[5]**

b) An input image has been converted into a matrix of size 16×16 along with a filter of size 3×3 with a Stride of 1. Determine the size of the convoluted matrix. **[5]**

c) What is the role of multiple Fully Connected (FC) Layers in CNN? Which variant of CNN uses multiple FC layers? **[5]**

